Solution Architecture Definition

Referee Solution –

Referee Management & Mentoring

v 0.4

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May 2026

Document History

Version No	Revision Date	Author	Summary of Changes
0.1	26-May-26	Shakkeeb Mohammed	Initial draft - ACB awareness
0.2	10-Aug-26	Shakkeeb Mohammed	Updated ADR decision - extend Referee service
0.3	11-Aug-26	Shakkeeb Mohammed	Pre-ACB review comments
0.4	14-Aug-26	Shakkeeb Mohammed	DAWG review comments

Document Review

Name	Role	Responsibility	Issue Date	Review Date
Shakkeeb Mohammed	Solution Architect	Responsible		
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Approvals

Name	Role	Approval	Approval Date
Data Understanding & Modelling Working Group	Data modelling approval	Approved	12-Aug-2026
Data Architecture Working Group	Data architecture approval	Approved	14-Aug-2026
Architecture Control Board	Delivery framework approval	Approved	20-Aug-2026

Reference Documents

Document	Link
PRD – View & manage referees	View & manage referees PRD.docx
PRD – Mentor management	Mentor management PRD.docx

Guidance

Solution Architecture Definition Document Guidance - Confluence

Reference the Solution Architecture Definition document guidance notes which defines the purpose of each section and suggests

- the type of information to include
- possible questions to answer
- types of diagrams to include

For any sections or sub-sections which are not relevant please retain the headings but state “Not applicable” or “Not required” with reasoning.

Table of Contents

1 Introduction

This Solution Architecture Definition (SAD) covers phase one of the Referee Solution initiative. In this phase, the scope is defined by two product requirement documents (PRDs), each addressing a key area of the overall solution.

- County FA Referee Development Officers currently **manage referee** details, eligibility, and status through retiring Whole Game System functionality. This essential capability supports the governance of over 30,000 referees and must be replaced in PFF to maintain service continuity.

  Future initiatives, including referee and mentor allocation, require referee management to move out of legacy systems and reduce reliance on outdated integrations. Referee development is also planned to transition into an FA shared service through Business Optimisation, which depends on a modern, centralised referee management capability.
- Referees **mentor management** is manual, inconsistent, and largely offline, with no centralised, safeguarding-compliant process to approve, assign, or monitor mentors. Counties rely on spreadsheets, emails, and informal communication, leading to unclear expectations, low mentor coverage, and limited assurance that mentoring is delivered safely by appropriately vetted individuals, in line with NSPCC Recommendation 8.

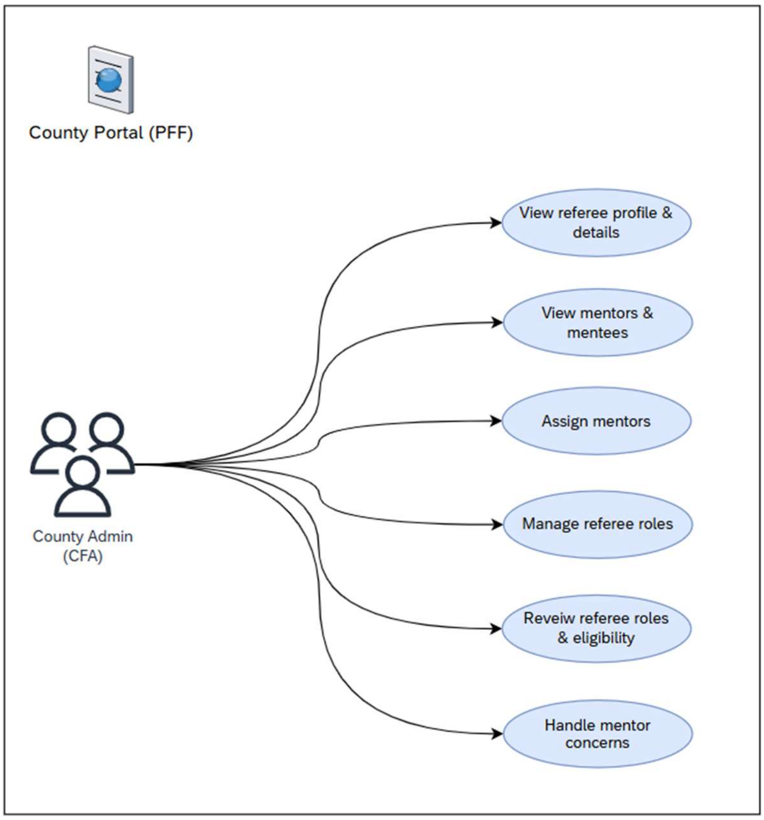
1.1 Objectives

The objectives of phase one are to replace key legacy capabilities, improve operational control, and establish a modern foundation for future referee services.

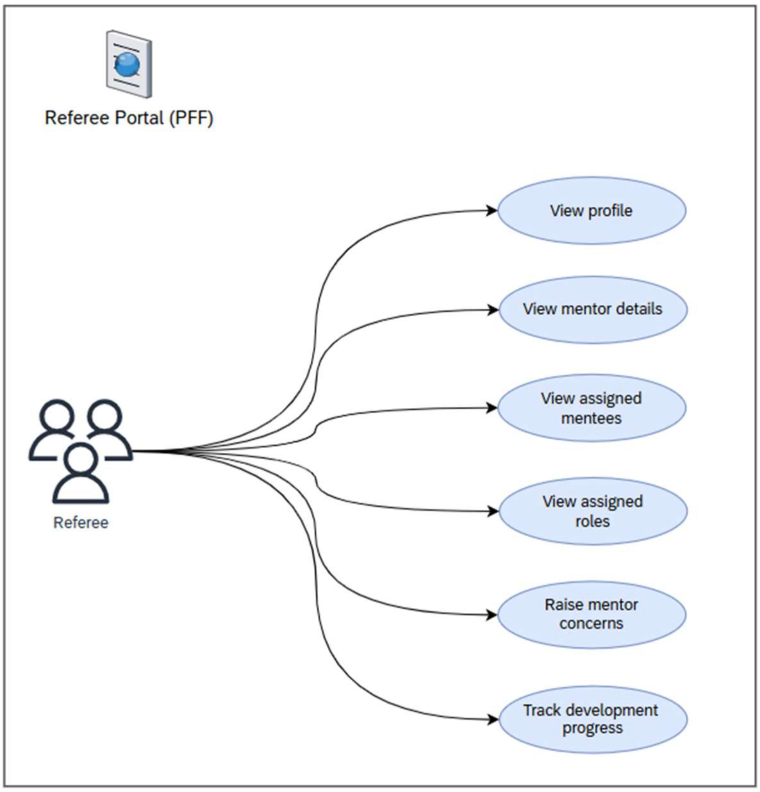
- Replace retiring Whole Game System referee management functionality in PFF to maintain service continuity for County FA operations.
- Provide a modern, centralised referee management capability that supports future initiatives, including referee and mentor allocation, and reduces reliance on legacy integrations.
- Introduce a consistent, safeguarding compliant mentor management process to improve approval, assignment, and oversight of grassroots mentoring.
- Establish the foundation for Business Optimisation and the planned transition of referee development into a shared service.

1.2 Functionality

1.2.1 County Portal



1.2.2 Referee Portal



1.3 Constraints

- **Qualification integration from LXP:** As the LXP target architecture is being developed in parallel, the solution must support referee qualification data directly from LXP rather than relying on IPS event data.

1.4 Assumptions

- **No integration back to WGS:** Referee data will not be integrated back into the Whole Game System as part of this phase.
- **Phased WGS decommissioning:** The solution should support the phased decommissioning of the WGS Referee module as equivalent PFF capabilities are delivered and operationally accepted.
- **Future Referee portal consolidation:** The Referee Registration Portal and Referee Portal will remain separate in this phase. Any future consolidation or functional merge will be considered separately as part of a later roadmap decision.
- **No Full-Time integration (except Referee info):** Safeguarding, qualification, DBS and suspension details will not be integrated from PFF to Full-Time in this phase.

1.5 Dependencies

Delivery of this phase depends on the availability and alignment of external platforms and integrations that provide key referee data and downstream consumption.

- **LXP qualification data:** The preferred approach is for LXP to provide referee qualification data in a PFF-consumable format aligned to the target architecture. However, this is not a delivery dependency, as the solution can still be delivered using the existing IPS data flow.
- **FullTime web services:** The solution depends on FullTime web services being available to receive and use referee information from PFF.
- **FullTime access and subject-matter expertise:** End-to-end validation of the referee flow between PFF and FullTime depends on timely access to FullTime environments and support from colleagues with relevant FullTime system knowledge.
- **MOAS access and subject-matter expertise:** End-to-end validation of the referee flow between PFF and MOAS depends on timely access to MOAS environments and support from colleagues with relevant MOAS system knowledge.
- **Referee reporting:** Delivery of referee reporting capabilities is dependent on the DSI platform and associated Power BI reporting solution being available.

1.6 Legacy

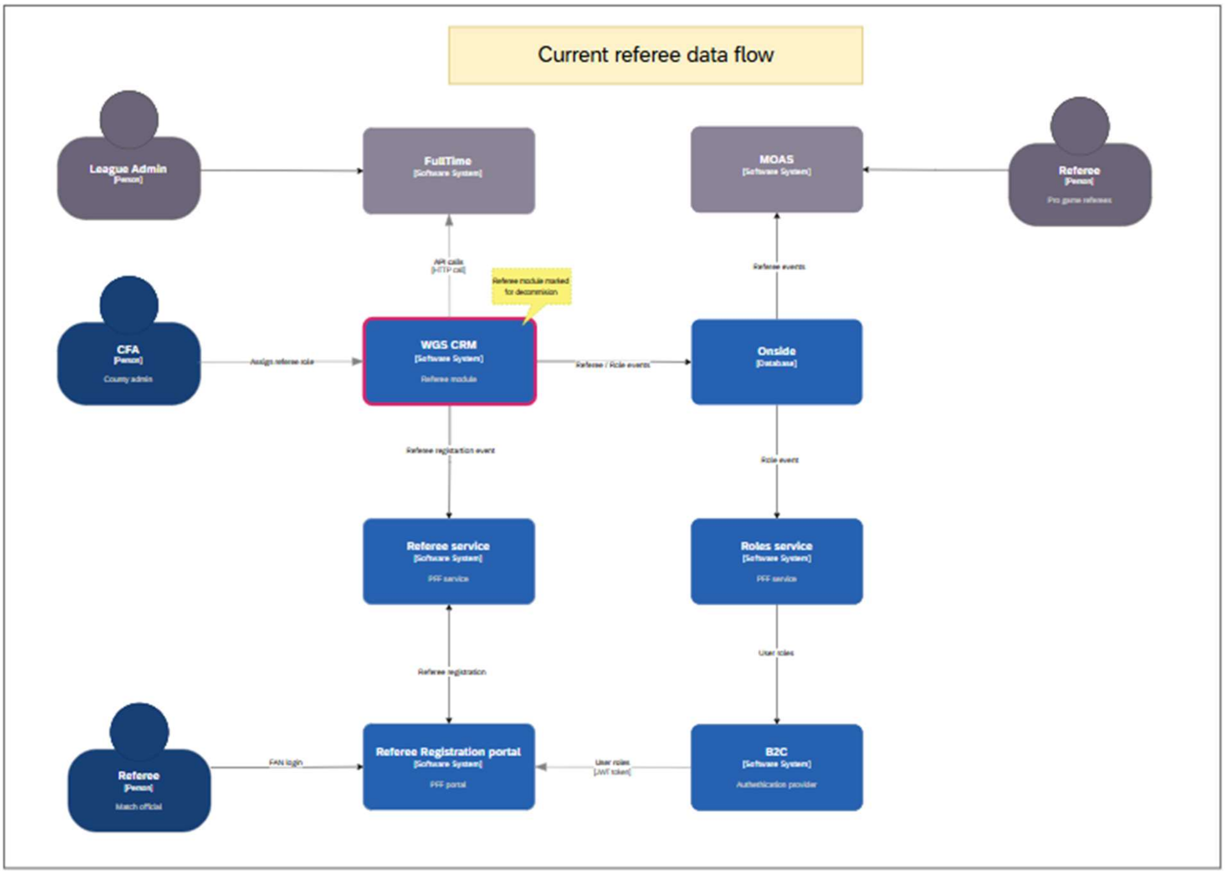
This phase is driven by the need to move key referee management capabilities away from legacy platforms and reduce dependency on outdated system integrations.

- Retirement of the WGS CRM Referee module
- Onside remains in scope for roles and referee information required for the MOAS flow.

2 Data

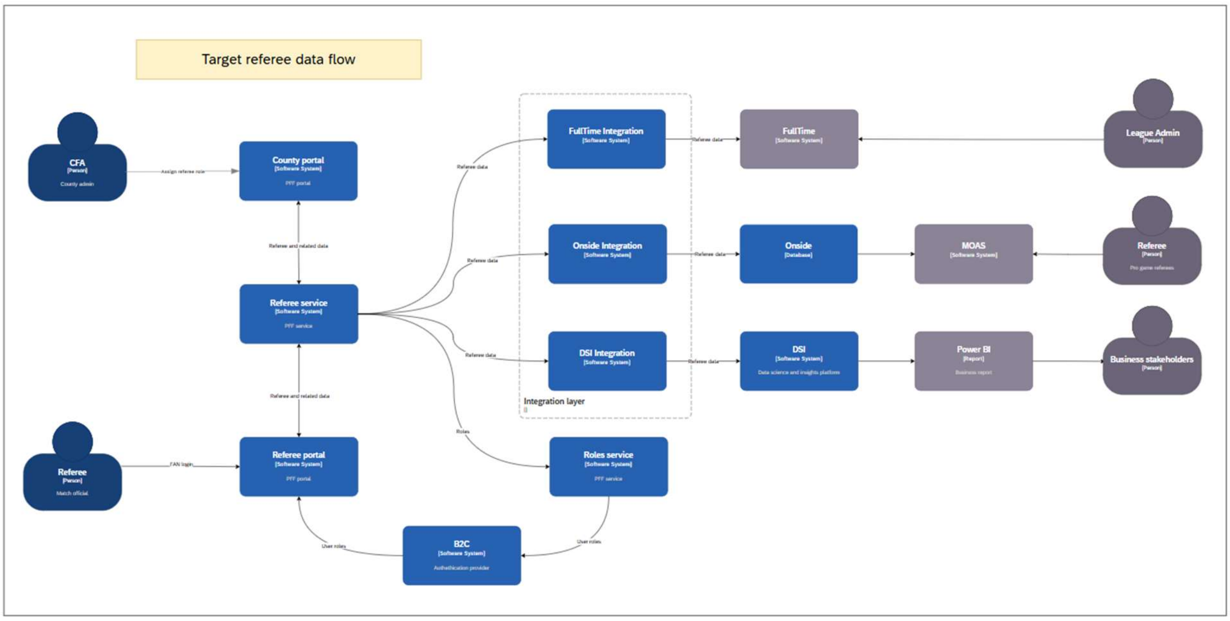
Current data flow

Referee-related data is currently mastered in WGS. FullTime does not support safeguarding verification during fixture assignment, and referee reporting is not yet available through DSI.



Target data flow

The target flow uses PFF as the master platform for referee management data, including mentoring. PFF will provide the Referee data required by FullTime and by DSI for reporting. And continue providing required to Onside database (for MOAS usage).



2.1 GDPR

There is no change to GDPR requirements for this phase. PFF standards are defined in the PFF Platform Architecture [R1].

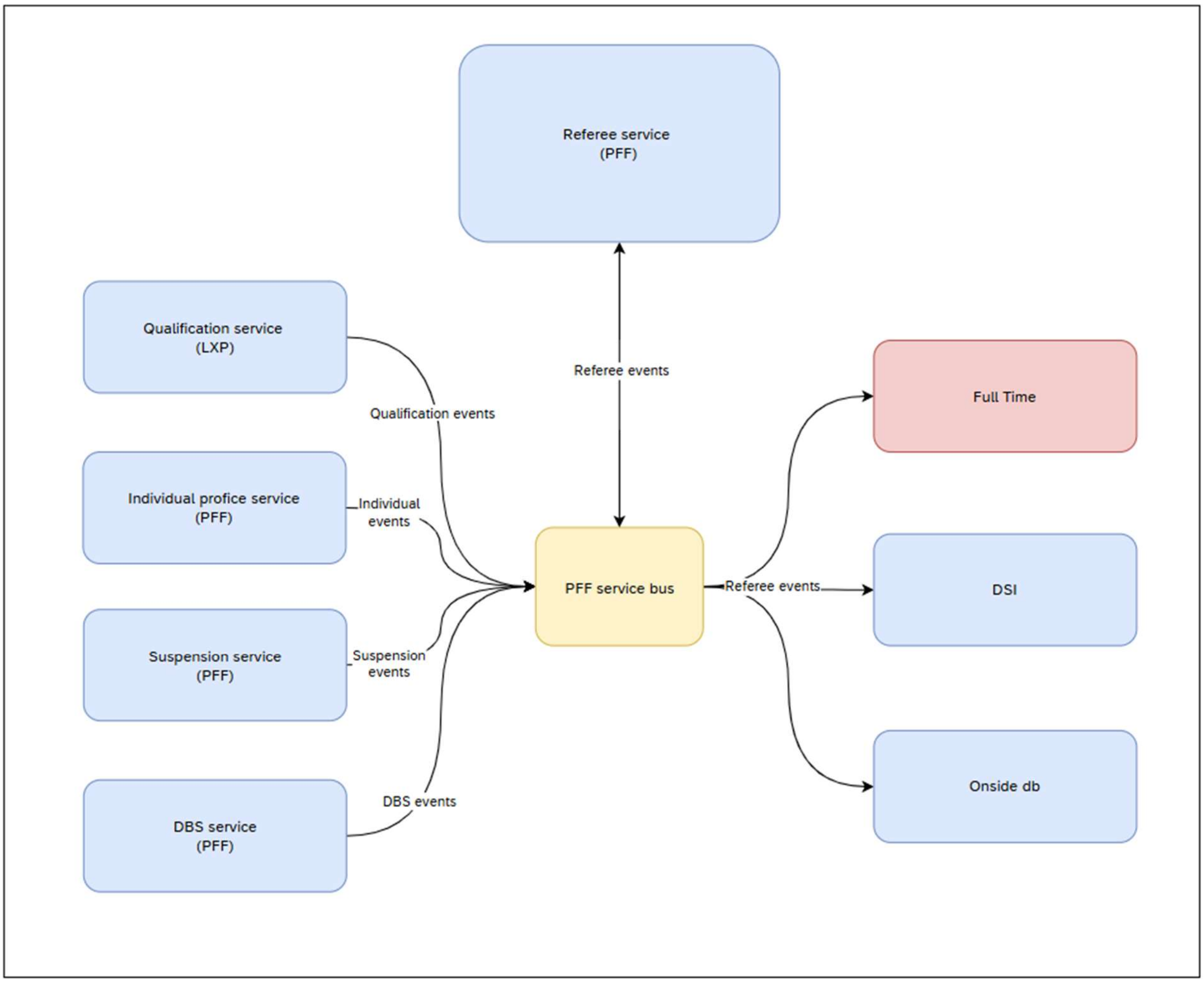
The following categories of personally identifiable information (PII) will be stored within The FA infrastructure:

- Referee and individual records
- Referee registration details
- Referee qualifications and suspension information
- Referee mentor details

2.2 Sources

- **Referee database** – Stores referee records and registration details.
- **Individual database** – Provides individual lookup information.
- **Suspension database** – Provides referee suspension information.
- **DBS applications database** – Provides DBS information.
- **LXP (Qualification database)** – Provides referee qualification data.

2.3 Integration



The integration architecture uses event-driven patterns to exchange referee-related data between PFF, source systems, downstream operational systems, and reporting platforms.

- **Source-system integrations:** Source systems publish individual, qualification, suspension, and DBS events for use by the Referee service.
- **Azure Service Bus:** Provides the event transport layer for reliable message delivery between services and integration components.
- **Referee service:** Subscribes to relevant events, processes the data, and stores it in the Referee database for operational use.
- **Downstream system integrations:** Downstream systems FullTime, DSI and Onside uses referee data published from the master source (Referee service)

Reference patterns

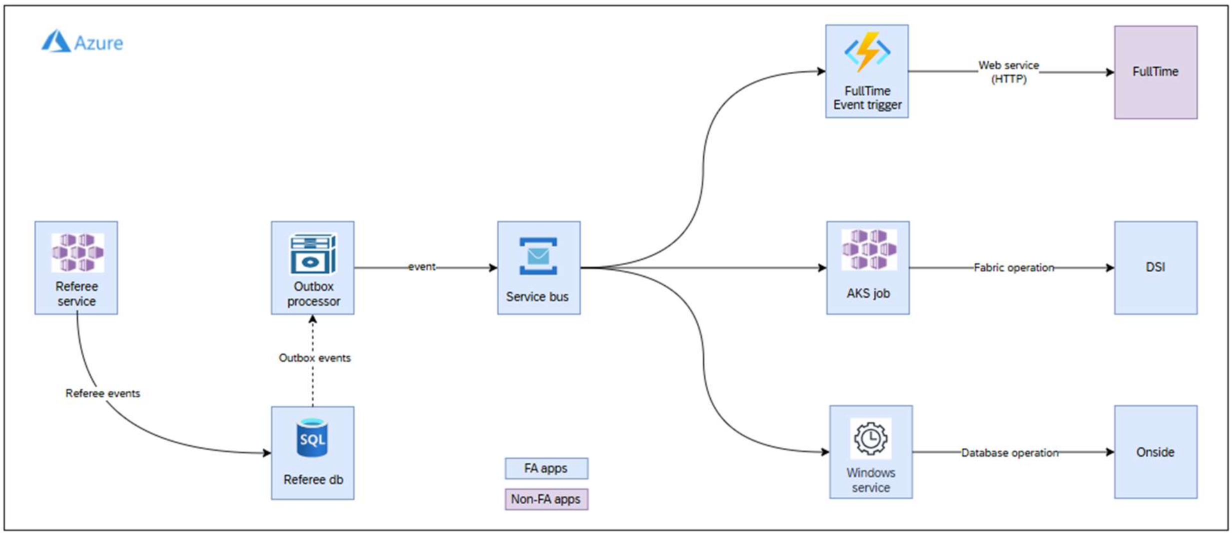
- Pub sub integration
- Transactional outbox pattern

2.3.1 Inbound Master Data Integration

Referee-related master data is received from the relevant source systems and stored in the Referee service for internal use. This includes individual, qualification, suspension, and DBS information required to support referee management within PFF.

- **Suspension data:** Referee suspension information is integrated from Suspension service.
- **DBS data:** DBS information is integrated from DBS service.
- **Individual data:** Individual data is integrated from Individual profile service.
- **Qualification data:** Referee qualification data is integrated from Individual profile service, until LXP target architecture is available. Once LXP is available, qualification data will be integrated from Qualification service.

2.3.2 Downstream Integration



FullTime Sync

FullTime Sync uses an Azure Function as an event-triggered integration wrapper. The function listens for referee-related events published to Azure Service Bus and, when an event is received, calls the appropriate FullTime web service to push the relevant referee data into FullTime.

DSI Sync

DSI Sync uses the RealTimeIngestion AKS job to support event-driven reporting integration. The AKS job listens for relevant PFF events and publishes them to Microsoft Fabric within the DSI platform, where they can be used for downstream analytics and reporting.

Onside Sync

Onside Sync is a Windows service that subscribes to referee-related events from the PFF Service Bus and updates the Onside database. This makes the relevant registration data available for downstream use by MOAS.

2.4 Migration

The solution requires a one-time migration of in-scope referee data into the Referee database to support Phase 1 delivery.

- **Source-system data:** Suspension data from the Suspension database, DBS data from the DbsApplications database, and individual and qualification data from the Individual Profile service will be migrated into the Referee database.
- **Legacy referee data:** WGS referee data will be migrated into PFF as part of the transition from the legacy referee management capability.

2.5 Audit

The solution will follow the standard PFF audit logging approach.

Audit data retention will be managed as follows:

- Only audit data for the current season will be retained in the live system.
- As part of the season rollover process, audit data from previous seasons will either be deleted or archived to the data lake, in line with the agreed retention approach.

https://the-fa.atlassian.net/wiki/x/GIDPxw

2.6 Backups

Backup approach

Backups for this solution will follow the existing PFF architecture defined in *"PFF Solution Architecture"* [R3].

2.7 Reporting

Reporting approach

Data from this solution will be ingested into the DSI platform to support current and future business reporting requirements.

3 Non-functional Requirements

3.1 Security

Security approach

Security for this solution will align with the existing PFF standard defined in *"PFF Platform Architecture"* [R1].

3.2 Capacity

Capacity approach

Phase 1 will use the existing PFF database elastic pool and Kubernetes infrastructure to support the required changes.

Key capacity assumptions for Phase 1 are as follows:

- The solution is expected to support approximately 30,000 referees per season.
- Document uploads are not expected to form part of the referee flow in this phase.

3.3 Performance

Key performance requirements for Phase 1 are as follows:

- Peak referee creation is expected to reach up to 500 records per day.
- New service APIs must meet a maximum response-time SLA of 3 seconds.

3.4 Scalability

Key scalability considerations for Phase 1 are as follows:

- The existing PFF API Management Gateway is manually scaled and is currently running on a single instance. No scaling required.
- The existing AKS infrastructure has been performance-tested to support up to 4,000 concurrent users.
- Azure Service Bus will use the Premium tier with two messaging units to handle the expected load.

Existing PFF AKS Cluster Capacity

- Node size: Standard E8s v4 (8 vcpus, 64 GiB memory)
- Node count: Min 3, Max 15

Recommended HPA settings for new services

- **CPU:** Request: 400, Limit: 800
- **POD count:** Min: 3, Max: 6

Existing RefReg databases will be renamed to Referee database, which already part of the existing PFF elastic pool.

3.5 Availability

Availability approach

Availability for this solution will align with the existing PFF standard defined in *"PFF Platform Architecture"* [R1].

3.6 Disaster Recovery

Disaster recovery approach

Disaster recovery for this solution will align with the existing PFF architecture defined in *"PFF Solution Architecture"* [R3].

3.7 Monitoring

Monitoring approach

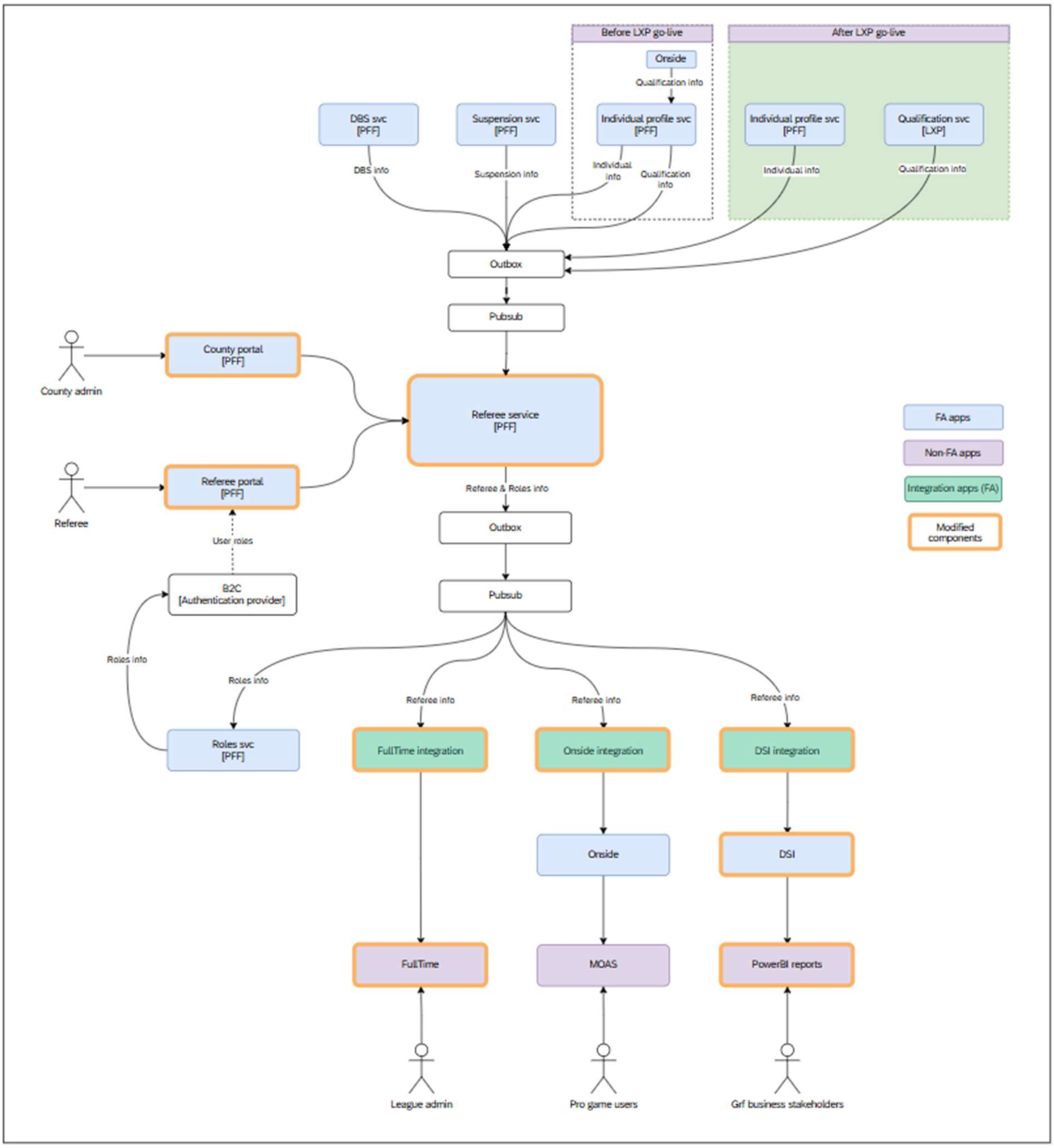
Monitoring for this solution will align with the existing PFF architecture defined in *"PFF Solution Architecture"* [R3].

4 Solution Architecture

This section describes the target solution architecture for the Referee Solution and explains how the application, infrastructure, environments, and DevOps capabilities work together to deliver the Phase 1 requirements. It provides a high-level view of the main solution components, their responsibilities, and the integration patterns used to support referee management, mentoring, reporting, and downstream system updates.

The architecture aligns with the existing PFF platform standards and reuses approved platform services wherever possible, including AKS, Azure SQL, Azure Service Bus, Azure Functions, API Management, monitoring, and deployment tooling. Solution-specific components are introduced only where required to support the new referee capabilities and integration requirements.

4.1 Application Architecture



As the Referee data is getting mastered in Referee service, all the applications show in the diagram are getting impacted in phase 1.

The application architecture is based on the existing PFF platform pattern and introduces solution-specific services only where required for Phase 1 referee capabilities. The key application components are summarised below:

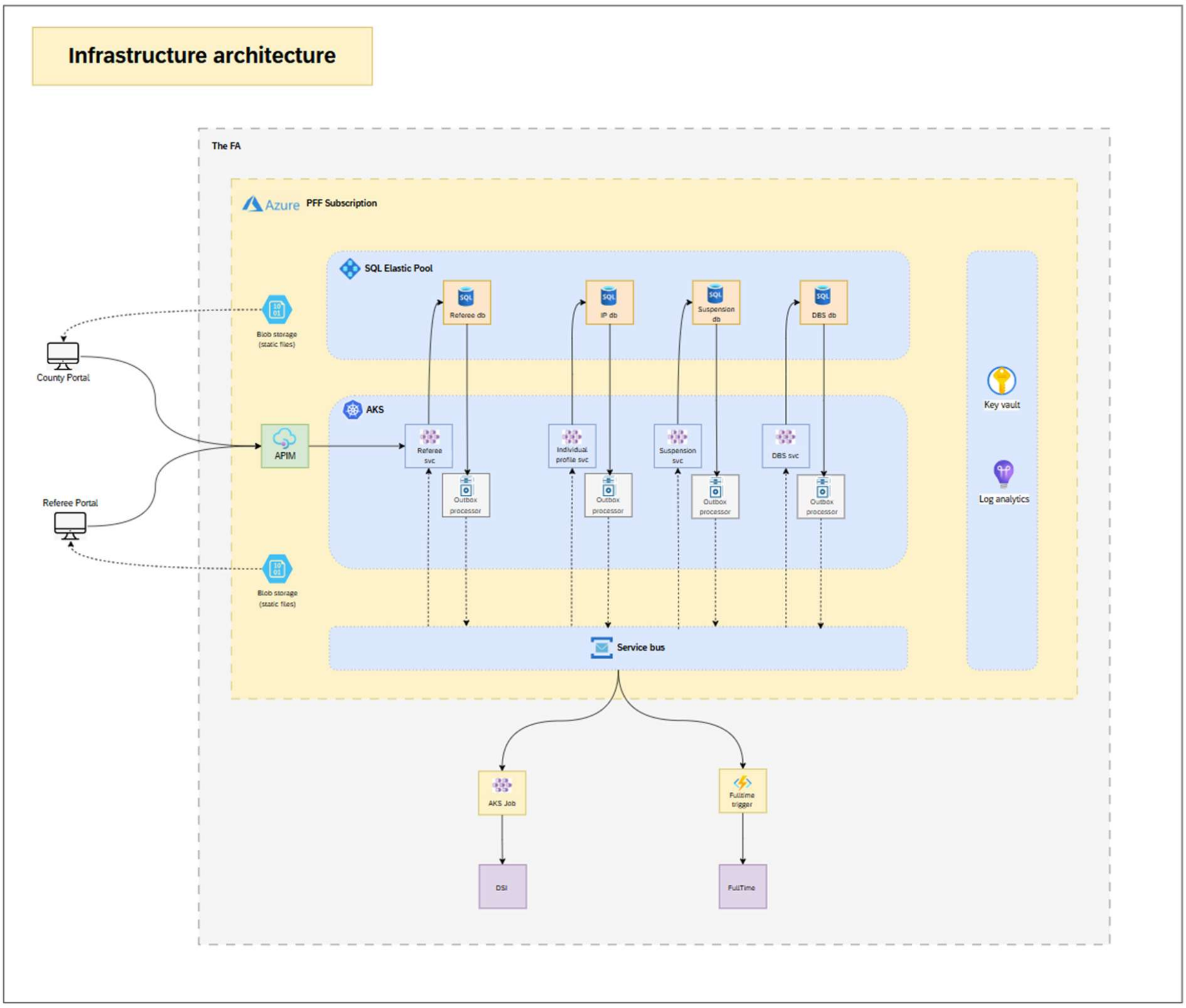
- **County Portal:** Provides the user interface for County FA users to view and manage referee and mentor-related information.
- **Referee Portal:** Provides the front-end capability for referees to complete registration-related activities and maintain information required for referee management within PFF.
- **Referee service:** Core business logic is written in PFF referee service, which interacts with pub sub to publish and consume events.
- **FullTime application:** Consumes referee details from Referee service for fixture appointment.
- **Pub sub:** Uses Azure Service Bus to publish and consume referee-related events between PFF services and downstream integration components.
- **Downstream integration services:** Azure Functions and jobs process relevant events and publish data to FullTime, Onside and the DSI platform for operational and reporting use cases.

4.1.1 Phased Delivery (LXP integration)

- **Before LXP go-live:** The Referee service will subscribe to both individual and qualification events from the Individual profile service.
- **After LXP go-live:** Qualification events will be subscribed from the Qualification service, while individual events will continue to be subscribed from the Individual profile service.

  LXP related changes are planned to address in Q3 of 2026-27 season.

4.2 Infrastructure Architecture



The infrastructure architecture reuses the existing PFF Azure platform wherever possible and introduces only the additional components required to support the Referee Solution. The key infrastructure components are summarised below:

- **Azure Kubernetes Service:** Hosts the PFF microservices that deliver the referee management and mentor management capabilities.
- **Azure API Management:** Provides the existing gateway layer for routing portal and integration traffic to the appropriate backend services.
- **Azure SQL Database:** Stores referee and mentor data in databases hosted within the existing PFF SQL elastic pool.
- **Azure Service Bus:** Supports event-driven communication between PFF services and integration components for referee-related data changes.
- **Azure Functions:** Provides lightweight integration processing for downstream syncs, including FullTime and DSI integration flows.
- **Key Vault:** Stores secrets, certificates, and configuration values required by the application and integration components.
- **Monitoring and logging:** Uses the existing PFF monitoring approach to capture application telemetry, operational logs, and integration diagnostics.

4.3 Environments

The solution will use the existing PFF and FullTime environment landscape. No new environments are expected to be required for Phase 1.

- **PFF environments:** Development, QA, Pre-Production, Load Test, and Production will be used for build, test, performance validation, and live operations.
- **Downstream systems environments:** Existing non-production and production environments will be used to support integration testing and live downstream synchronisation.

4.4 DevOps

The solution will use the existing PFF DevOps delivery model. No new DevOps tooling, pipelines, or release processes are required for Phase 1.

- **Build and deployment:** Existing PFF pipelines will be used to build, package, and deploy the relevant services.
- **Release process:** Releases will follow the current PFF promotion and approval process across environments.
- **Operational alignment:** Deployment, monitoring, and rollback activities will remain aligned to established PFF platform practices.

4.5 ADR Logs

Reference	Outcome
ADR15: Referee solution - Extend Referee service	• Expand the existing Referee service rather than introducing a new microservice. • Maintain clean isolation through namespaces, modular classes, folders and separate schema. • Rename RefReg database to Referee database.

5 Technology Stack

The technology stack for Phase 1 builds on the existing PFF platform and uses approved cloud, data, integration, and development technologies wherever possible. The solution will reuse established PFF infrastructure services, delivery tooling, monitoring, and operational patterns, with solution-specific components introduced only where required to support referee management, mentor management, reporting, and downstream integration capabilities.

5.1 Infrastructure

The core infrastructure technologies for this solution are listed below:

- Azure Kubernetes Service (AKS)
- Azure API Management
- Azure AD B2C
- Azure Functions
- Azure Service Bus
- Azure log analytics
- Azure key vault

5.2 Data

- Azure SQL Database with Elastic Pool

5.3 Development

The core development technologies and delivery tools for this solution are listed below:

- Microsoft Visual Studio 2026 or Visual Studio Code
- .NET 10, Entity Framework 10, and C# 14 for microservices
- Angular 21 for the front-end portal
- Azure SDK
- Azure DevOps repositories, pipelines, and release tools

6 Costs

6.1 Infrastructure

Cost Area	Approach / Usage	Cost Impact
Portals and CDN profiles	No new portals or CDN profiles are required.	No extra cost
Azure SQL Database	No new database	No extra cost
Microservices	No new micro service	No extra cost
API Management	The existing PFF API Management instance will be used.	No extra cost
Service Bus	The existing PFF Service Bus will be used.	No extra cost
Key Vault	The existing PFF Key Vault will be used for secrets and configuration.	No extra cost
Microsoft Fabric / DSI	DSI data ingestion cost	Low

6.2 Software

Software licensing approach

No additional software licensing or third-party SaaS products are required for Phase 1.

- The solution will use existing approved platform services and tooling.
- There are no expected incremental software subscription costs for this phase.

7 Maintenance

7.1 Support

Support approach

Support for this solution will follow the standard PFF support model, using the existing operational processes, support channels, and escalation routes.

- Operational support will be managed through the established PFF support process.
- Incidents and service requests will follow the agreed triage, prioritisation, and escalation procedures.
- Application and platform teams will provide support in line with their existing responsibilities.

7.2 Roadmap

Roadmap overview

This document covers Phase 1 of a two-phase delivery roadmap for the Referee Solution.

- **Phase 1:** Focuses on the foundational referee management capabilities described in this document.
- **Phase 2:** Will extend the solution to cover referee availability and appointments.

7.3 Life Expectancy

Life expectancy and decommissioning approach

As Referee functionality is transitioned from WGS to PFF, the equivalent WGS capabilities and related integrations will be progressively disabled to avoid duplication and reduce legacy dependency.

- PFF will become the target platform for the referee management functionality delivered in this phase. The lifecycle of this solution is defined as five years from the implementation date. This should be reviewed and revisited at a later stage.
- WGS functionality will be disabled once the equivalent PFF capability is live, validated, and operationally accepted.
- Legacy integrations from WGS will be retired where they are replaced by PFF integrations or are no longer required.

7.4 Decommission

The decommissioning approach will ensure that retired WGS referee management capabilities are removed in a controlled manner once the equivalent PFF capability is live, validated, and operationally accepted.

- **Capability retirement:** WGS referee management functionality will be disabled after the corresponding PFF capability has been deployed, tested.
- **Integration retirement:** Legacy WGS integrations will be retired where PFF becomes the agreed source of referee data for downstream systems.
- **Data retention and access:** Any historical referee data required for audit, reporting, or operational reference will be retained line with agreed data retention.
- **Operational readiness:** Support teams, users, and downstream system owners must be informed before decommissioning activity is completed.

8 Appendix

8.1 References

#	Name	Reference
1	PFF Platform Architecture	PFF Platform Architecture.docx
2	PFF Non-Functional Requirements	PFF Non-Functional Requirements.xlsx
3	PFF Solution Architecture	PFF Solution Architecture
4	LeanIX Diagram Reference	LeanIX Diagram Reference